PROTEXXA CYBER NATIONS SCOTIARISE
CYBER SECURITY ANALYST PROGRAM
ASSIGNMENT 4: NETWORKING

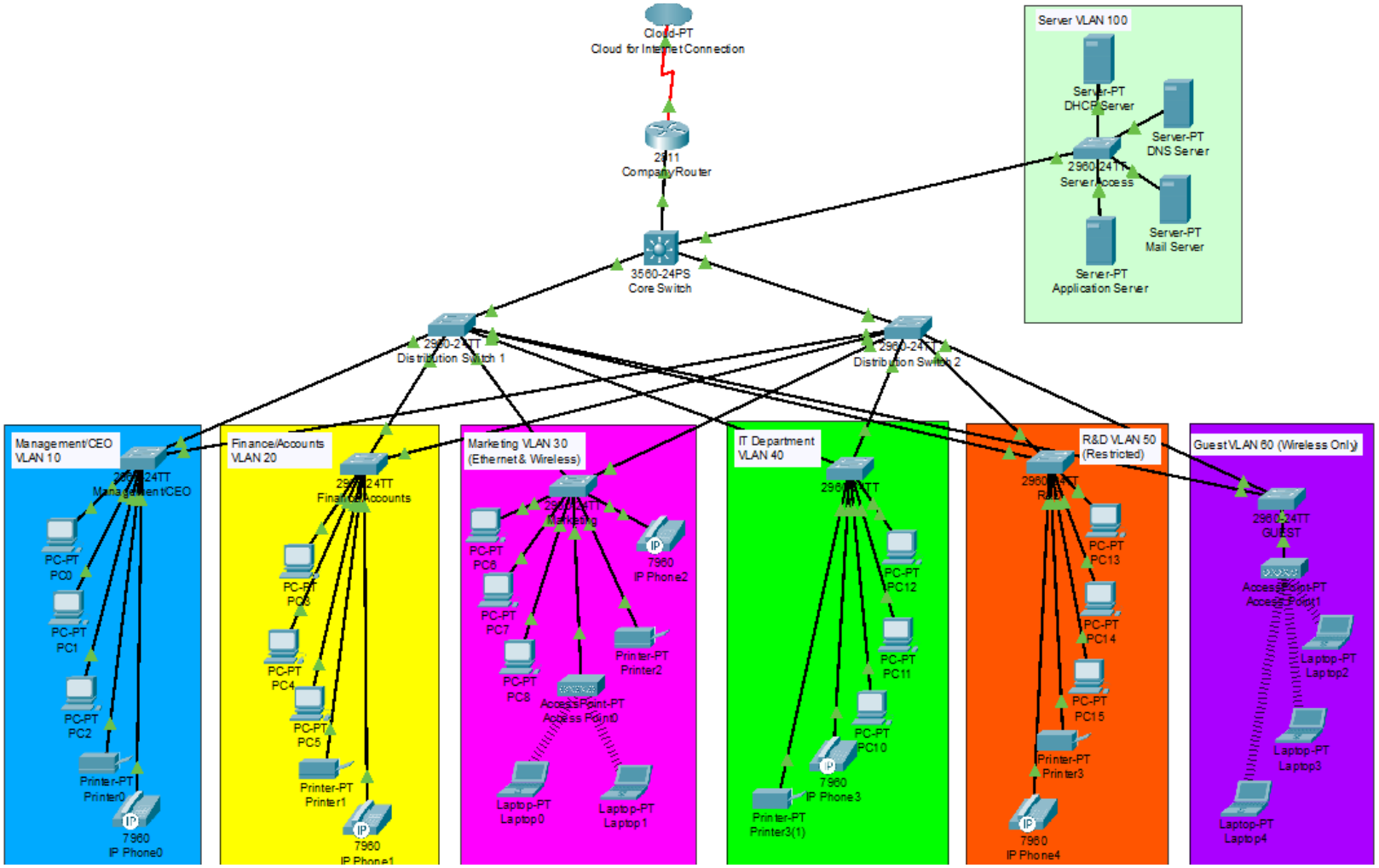
NETWORKING

SECURE ENTERPRISE NETWORK INFRASTRUCTURE DESIGN WITH HIERARCHICAL VLAN ARCHITECTURE WITH ADVANCED SECURITY IMPLEMENTATION



Zolian R. Smith
July 28, 2025

Network Topology



Network Topology Description and Justification

Topological Choice and Structure

The network design implements a hierarchical three-tier architecture consisting of:

1. **Core Layer**: Single core switch (3560-24PS) connected to the company router for internet connectivity
2. **Distribution Layer**: Two distribution switches (Fa0/7 Distribution Switch 1 and Fa0/7 Distribution Switch 2) providing departmental connectivity
3. **Access Layer**: Individual department switches within each VLAN providing end-device connectivity

This topology utilizes VLAN segmentation with the following structure:

- **Management/CEO:** VLAN 10 (192.168.10.0/27)
- **Finance/Accounts:** VLAN 20 (192.168.20.0/27)
- **Marketing:** VLAN 30 (192.168.30.0/26) - larger subnet for ethernet & wireless
- **IT Department**: VLAN 40 (192.168.40.0/27)
- **R&D:** VLAN 50 (192.168.50.0/27)
- **Guest:** VLAN 60 (192.168.60.0/27) - wireless only
- **Server VLAN**: VLAN 100 (192.168.100.0/27)

Connection Status Documentation

All Connections are Live (Green):

- **Core switch to Router**: Trunk connection successfully carrying all VLAN traffic with proper 802.1Q tagging
- **Core switch to Distribution switches**: Trunk connections enabling seamless multi-VLAN routing
- **Distribution switches to department access switches**: All VLAN assignments properly configured and operational
- **End devices to access switches**: All PCs, printers, and IP phones successfully connected to their respective VLANs
- **Server connections**: All servers in VLAN 100 are live and accessible
- **Wireless Access Points**: All wireless connections for Marketing and Guest VLANs are operational

Network Status:

The network has achieved full convergence with all links operational, demonstrating proper VLAN configuration, trunk/access port assignments, and successful inter-VLAN routing through the core infrastructure. This design ensures proper VLAN traffic flow while maintaining security boundaries and enabling controlled inter-department communication through the router's sub-interfaces.

How the Topology Fulfills the Four Network Design Requirements

1. **Segmentation:** VLANs isolate each department with dedicated subnets, containing broadcast domains and enabling firewall rule enforcement. Sensitive departments (R&D, Finance) remain completely isolated from guest networks through Layer 2 VLAN tagging.
2. **Scalability:** The hierarchical three-tier architecture allows easy expansion. New departments can be added as VLANs without reconfiguration, distribution switches accommodate growth, and subnet sizing provides room for additional hosts (30+ per department, 62 for Marketing).
3. **Single Points of Failure Mitigation:** Dual distribution switches provide redundant paths between access and core layers. Multiple uplinks ensure path redundancy, while the dedicated Server VLAN maintains critical service availability even during individual switch failures.
4. **Server Placement:** All servers are centralized in Server VLAN 100 for optimal security, management, and performance. This placement enables granular access control through firewall rules while providing necessary services (DHCP, DNS, Mail, Application) to authorized departments only.

Q1: Firewall rules

Rules		Implementation
i.	**Deny pings from the internet:**	`access-list 101 deny icmp any 192.168.0.0 0.0.255.255 echo`
ii.	**Deny mail leaving R&D subnet:**	`access-list 102 deny tcp 192.168.50.0 0.0.0.31 any eq 25`
iii.	**Deny requests from internet to R&D:**	`access-list 103 deny ip any 192.168.50.0 0.0.0.31`
iv.	Suggested rule: **Block Guest VLAN from accessing internal servers**	`access-list 104 deny ip 192.168.60.0 0.0.0.31 192.168.100.0 0.0.0.31`
v.	Suggested rule: **Block R&D from accessing Finance subnet**	`access-list 105 deny ip 192.168.50.0 0.0.0.31 192.168.20.0 0.0.0.31`
vi.	Suggested rule: **Block Guest from accessing Management subnet**	`access-list 106 deny ip 192.168.60.0 0.0.0.31 192.168.10.0 0.0.0.31`
vii.	Suggested rule: **Block SSH access from internet**	`access-list 107 deny tcp any any eq 22`
viii.	Suggested rule: **Block FTP access from internet**	`access-list 108 deny tcp any any eq 21`
ix.	Suggested rule: **Block Telnet access from internet**	`access-list 109 deny tcp any any eq 23`
x.	Suggested rule: **Block Marketing from accessing R&D subnet**	`access-list 110 deny ip 192.168.30.0 0.0.0.63 192.168.50.0 0.0.0.31`

Justification for 3 mandatory rules:

a. **Deny pings from internet (3 lines)**

This rule prevents external attackers from using ICMP ping requests to discover active hosts on the internal network. Ping sweeps are commonly used in reconnaissance attacks to map network topology. Blocking these requests reduces the network's visibility to potential threats and prevents information disclosure about internal infrastructure.

b. **Deny mail leaving R&D (3 lines)**

This rule prevents sensitive research and development information from being accidentally or maliciously transmitted via email outside the organization. R&D departments typically handle confidential intellectual property, trade secrets, and proprietary technologies. Blocking outbound email ensures that sensitive data remains within the organization and prevents data exfiltration through email channels.

c. **Deny requests from internet to R&D (3 lines)**

This rule creates a security perimeter around the R&D subnet by blocking all incoming traffic from the internet. R&D networks contain highly sensitive information that requires maximum protection from external threats. This rule prevents direct attacks, unauthorized access attempts, and potential data breaches by ensuring the R&D subnet remains completely isolated from external networks.

Q2: Departments without IP phone/printer

Department 1 without an IP phone nor a printer: Guest Department

Reason:

Guest users are temporary visitors who only require basic internet access and should not have access to corporate communication systems or printing facilities for security and cost management purposes.

Department 2 without an IP phone nor a printer: Server VLAN

Reason:

The server VLAN contains infrastructure servers (DHCP, DNS, Mail, Application) that provide network services and do not require user-facing communication devices like phones or printers.

Q3: Subnetting Calculations for /27

Number of subnets for /27 network: **8**

Number of usable hosts on each subnet: **30**

Network ID address	First usable host IP address	Last usable host IP address	Broadcast IP address
192.168.10.0	192.168.10.1	192.168.10.30	192.168.10.31
192.168.10.32	192.168.10.33	192.168.10.62	192.168.10.63
192.168.10.64	192.168.10.65	192.168.10.94	192.168.10.95
192.168.10.96	192.168.10.97	192.168.10.126	192.168.10.127
192.168.10.128	192.168.10.129	192.168.10.158	192.168.10.159

Q4: Strategies for Inter-Department Communication

Strategy 1: Access Control Lists (ACLs) with Permit Rules

Configure specific ACL rules that allow necessary communication between departments while maintaining security. For example, permit HR to access payroll servers in Finance, or allow IT department to access all subnets for maintenance purposes. This provides granular control over which departments can communicate and what services they can access, ensuring business operations continue while maintaining security boundaries.

Strategy 2: DMZ (Demilitarized Zone) for Shared Resources

Create a separate VLAN/subnet for shared resources like file servers, shared printers, and collaboration tools that multiple departments need to access. This DMZ acts as a neutral zone where departments can access common resources without directly communicating with each other's internal networks. This approach maintains departmental isolation while providing controlled access to necessary shared services.

Strategy 3: Time-Based and Role-Based Access Controls

Implement access controls that allow inter-department communication based on user roles and time schedules. For instance, managers from different departments can communicate during business hours, or emergency protocols can temporarily open communication channels during crisis situations. This strategy provides flexibility while maintaining normal security postures and can be automated based on business requirements.

Q5: Switch Port Type

Switch port type connected to router: Trunk Port

Reason why this switch port type is used:

A trunk port is used to connect the Core Switch to the router because it needs to carry traffic for multiple VLANs (10, 20, 30, 40, 50, 60, 100) simultaneously. The trunk port uses 802.1Q VLAN tagging to identify which VLAN each frame belongs to, allowing the router to perform inter-VLAN routing with its sub-interfaces. An access port would only carry traffic for a single VLAN, which would not support the multi-VLAN architecture required for departmental segmentation. The trunk configuration enables the router to act as the default gateway for all VLANs while maintaining proper network segmentation and security.